

Designation: C 2789-24

Standard Specification for Advanced Ceramic Matrix Composite (CMC) Turbine Components

This standard was issued on April 8, 2024; the version of record is continuously maintained.

1. Scope

1.1 This specification covers silicon carbide fiber reinforced ceramic matrix composite components for high-temperature turbine applications

1.2 The ceramic matrix composite turbine components covered by this specification are intended for use in gas turbine engines, aerospace propulsion systems, and power generation equipment operating at temperatures exceeding 1200°C.

1.3 The values stated in SI units are to be regarded as standard. No other units of measurement are included in this standard.

1.4 This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use.

2. Referenced Documents

2.1 ASTM Standards:

C 1275 Test Method for Monotonic Tensile Behavior of Continuous Fiber-Reinforced Advanced Ceramics

C 1358 Test Method for Monotonic Compressive Strength Testing of Continuous Fiber-Reinforced Advanced Ceramics

C 1366 Test Method for Tensile Strength of Monolithic Advanced Ceramics at Elevated Temperatures

E 1461 Test Method for Thermal Diffusivity by the Flash Method

3. Terminology

3.1 Definitions:

3.1.1 ceramic matrix composite—a composite material consisting of ceramic fibers embedded in a ceramic matrix

3.1.2 environmental barrier coating—a protective coating applied to prevent oxidation and degradation in combustion environments

3.1.3 interphase—a thin coating applied to fibers to control fiber-matrix bonding and enable toughening mechanisms

4. Classification

4.1 ceramic matrix composite turbine components shall be classified according to fiber architecture, matrix composition, and maximum service temperature.

- 4.1.1 **Class 1**—2D woven SiC/SiC, service temperature up to 1200°C
- 4.1.2 **Class 2**—2.5D woven SiC/SiC, service temperature up to 1400°C
- 4.1.3 **Class 3**—3D woven SiC/SiC, service temperature up to 1500°C

5. Ordering Information

5.1 Orders for material under this specification shall include the following information:

- 5.1.1 ASTM designation and year of issue
- 5.1.2 Class (1, 2, or 3)
- 5.1.3 Component type and geometry
- 5.1.4 Quantity
- 5.1.5 Environmental barrier coating specification (if required)
- 5.1.6 Quality assurance level

6. Materials and Manufacture

- 6.1 Reinforcement shall be continuous silicon carbide fibers with diameter 10-15 μm .
- 6.2 Matrix shall be silicon carbide deposited by chemical vapor infiltration (CVI) or polymer infiltration and pyrolysis (PIP).
- 6.3 Fiber coating (interphase) shall be boron nitride or carbon with thickness 0.1-0.5 μm .

7. Chemical Composition

7.1 The material shall conform to the chemical composition requirements specified in Table 1.

Element	Composition, %
Silicon carbide (fiber + matrix)	$\geq 85\%$
Boron nitride (interphase)	2-5%
Free silicon	$< 5\%$
Oxygen	$< 2\%$
Porosity	10-20%

8. Mechanical and Physical Requirements

Property	Requirement	Test Method
Tensile strength (RT)	≥ 200 MPa	ASTM C 1275
Tensile strength (1200°C)	≥ 150 MPa	ASTM C 1366
Elastic modulus	200-300 GPa	ASTM C 1275
Thermal conductivity	15-25 W/m·K	ASTM E 1461

9. Dimensions and Permissible Variations

9.1 Dimensional tolerances shall be specified on component drawings.

9.2 General tolerance on machined surfaces shall be ± 0.25 mm unless otherwise specified.

9.3 Surface roughness shall be $Ra \leq 3.2$ μm on critical surfaces.

10. Workmanship, Finish, and Appearance

10.1 Components shall be free from cracks, large pores, and delaminations detectable by non-destructive evaluation. Minor surface defects may be acceptable if they do not affect structural integrity.

11. Sampling

11.1 Qualification specimens representing each component geometry shall be manufactured and tested. Production components shall undergo 100% non-destructive inspection.

12. Test Methods

12.1 **Mechanical Testing**—Test representative specimens at room temperature and elevated temperatures per ASTM C 1275 and C 1366.

12.2 **Microstructural Analysis**—Examine polished cross-sections using optical and scanning electron microscopy.

12.3 **Non-Destructive Evaluation**—Inspect all components using ultrasonic, radiographic, or computed tomography methods.

13. Inspection

13.1 Inspection of the material shall be made as agreed upon between the purchaser and the seller as part of the purchase contract.

14. Rejection and Rehearing

14.1 Material that fails to conform to the requirements of this specification may be rejected. Rejection should be reported to the producer or supplier promptly and in writing.

15. Certification

15.1 A manufacturer's or supplier's certification shall be furnished to the purchaser that the material was manufactured, sampled, tested, and inspected in accordance with this specification and has met the requirements.

16. Product Marking

16.1 Components shall be marked with unique serial numbers using laser engraving or other permanent methods that do not compromise structural integrity.

17. Packaging and Package Marking

17.1 Components shall be individually packaged in custom foam inserts within rigid containers. Containers shall be sealed and labeled with handling instructions.

18. Keywords

18.1 ceramic matrix composite; CMC; silicon carbide; turbine; high temperature; aerospace